

7824 Zero Road, second floor

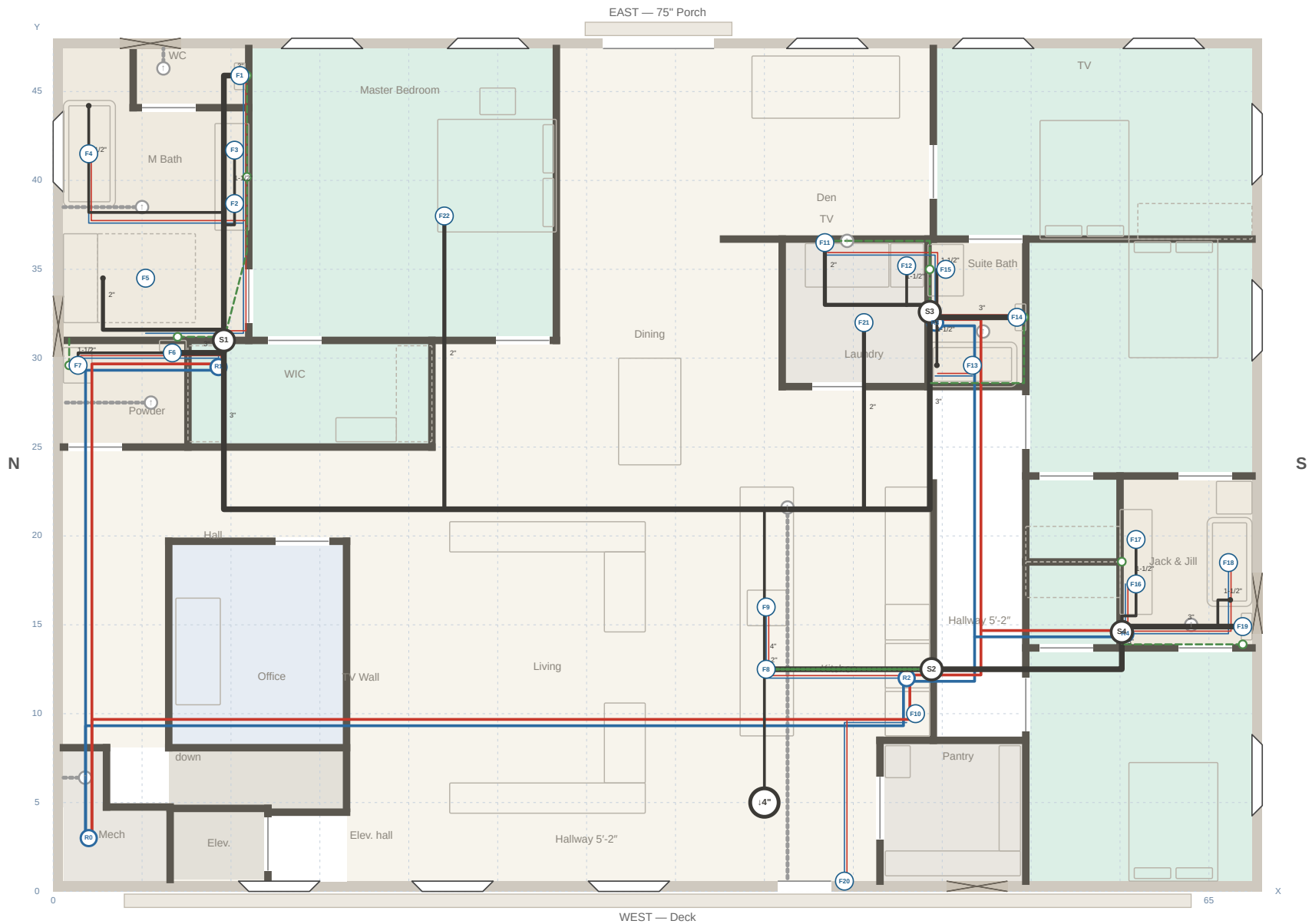
P-1 Plumbing plan · packet for the plumbing contractor

Fixtures, drains to the single drop, vent stacks, supply trunks and branches, and the exhaust ducts for coordination.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example P-1 S3).

Contents

1. P-1 sheet
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— drain under the floor (3" / 2" / 1-1/2" by weight)

- - - vent in wall or ceiling space

— cold / hot 3/4" trunk, 1/2" branches

..... exhaust duct

7824 Zero Road — second floor

Casper, Wyoming · 68' x 48' to brick, faces 7" inside · 9' ceiling · grid 5'
Wall numbers as the floor plan drawing of 2026-09-29

Plumbing plan

Fixtures, one drop, vent stacks, supply trunks and exhaust ducts for coordination
Schematic for pricing and coordination, not a stamped construction document. Verify on site

P-1

rev C · 2026-10-01
drawn for K. Christianson

Notes

1. Positions are taped from the nearest wall framing, exterior walls first. Interior walls: to the face of the studs; W# is the wall number on S-1. Exterior walls: to the face of the exterior studs as framed, 6" in from the brick. "On W12" means the item sits in that wall, and the distance runs along it. Wet walls are dimensioned to their 2x4 line: on the side a 2x6 or 2x10 grows toward, its face sits 1-1/2" (2x10: 5-1/4") closer.
2. Code basis: IRC 2021 chapters 25–31 (confirm the edition adopted by Casper / Natrona County). Sizes are minimums: toilets 3" with 3" closet bends, showers 2", tubs, lavs, sinks 1-1/2" (1-1/4" lav traps), washer standpipe 2" with a 2" trap 18"–42" AFF.
3. One drop. Every drain on the floor runs inside the open-web trusses to the drop at 4'-6" from west ext. wall, 6'-4" from W14 (hallway by the deck door), where a 4" line goes through the floor. Mains: M1 3" from the master group, M2 3" from the laundry group, M4 3" from the Jack & Jill group, M3 4" from the junction at 21'-0" from west ext. wall, 39'-6" from north ext. wall to the drop. S1 to the drop is 56': at the 1/8" per foot allowed for 3" and larger (IRC P3005.3) that is 7" of fall, at 1/4" per foot 14"; set M1 high in the truss space and hold 1/4" per foot where the depth allows. Fixture branches (2" and smaller) stay at 1/4" per foot.
4. Vents: S1 to S4 are now vent stacks. Each rises in the wall named from the base of its group (3" at S1, 2" at the others) and the branch vents join it in the ceiling space at least 6" above the highest flood rim. S3 and S4 may join to one 3" VTR to save a roof penetration.
5. Floor drains: F21 in the laundry and F22 in the master bedroom at 6'-2" from W19, 9'-6" from east ext. wall, 2" with trap primers (F21 from the washer box, F22 from the nearest lav). F22 lands under the foot of the bed as drawn; the floor around it slopes 1/8" per foot for 3'.
6. Island sink F8: loop vent per IRC P3112, hidden: the 2" vent rises inside the island to the counter underside, loops back down and runs under the floor (dotted green) to wall 16, where it rises as vent stack S2. Nothing goes up through the ceiling at the island. An air-admittance valve under the island is the fallback where the inspector allows it (P3114).
7. Cooktop downdraft (E-1): blower in the island base, 6" duct in the floor west to a wall cap on the west wall below the deck. The cap must stay clear of the deck framing; a roof-side route is the alternative.
8. Roll-in shower F5: 2" linear drain along the bench side (2'-0" from the north ext. wall), floor recessed 2" and sloped 1/4" per foot, bonded waterproofing to 6' on the wet walls. Pressure-balance / thermostatic valve on wall 20 at 42" AFF, hand shower on a slide bar reachable from the bench (ANSI A117.1 608).
9. No water piping in the exterior furring (Wyoming winters). Powder lav F7 and the master tub F4 are on exterior walls: supply them from below through the floor (floor-mounted tub filler; lav supplies up through the cabinet floor). Only the F7 vent uses the furring.
10. Supply: 1" cold and 1" hot up from the heater in the mech closet (R0), 3/4" PEX-A trunks under the floor (TA west, TB east, TC Jack & Jill), 1/2" branches. About 32 WSFU on this floor. Hot recirculation: 1/2" return from R4 and R1 back to the heater on a demand pump; insulate all hot and recirc lines. Stop valves at every fixture; a floor-level shutoff for the whole floor in the mech closet.
11. Accessible fixtures: wall-hung roll-under vanities with insulated P-traps and supplies (27" knee clear); comfort-height toilets 17"–19" with bidet-seat power (see E-1); lever or touch faucets; the suite and Jack & Jill tubs get a hand shower and pressure-balance valve.
12. Exhaust (for coordination, not plumbing scope): bath fans ducted to wall caps or the roof as drawn in grey, dryer 4" rigid through the roof (clean-out at the base), hood 6" through the roof with a backdraft damper.
13. The elevator sump / pit drain, the first-floor stack continuations and the water-heater sizing are first-floor items; the four stack locations and the R0 riser are the hand-off points.

Open items

- The drop at 4'-6" from west ext. wall, 6'-4" from W14: confirm the chase below and that a 4" line fits. Main M3 runs under the island and the living room floor; the truss manufacturer needs the route for chord and web clearance.
- F22 at 6'-2" from W19, 9'-6" from east ext. wall sits under the master bed's foot. Say if it should move clear of the bed.
- Hot water comes from B-1, the Navien combi boiler on hydronic sheet M-3, mixed to 120°F at TMV-1. That sheet has no recirculation line: keep the loop drawn here (recommended: the Jack & Jill is 70' of pipe from the mech closet), and if so, where does its return tie in at B-1?
- Tub choices: freestanding master tub with a floor filler as drawn, or a deck-mount with a 2x6 end wall; alcove tub/showers in the suite and Jack & Jill.
- Whether the kitchen gets a second sink or a pot filler (the cooktop is on wall 16; a pot filler is an easy 1/2" branch from R2).
- Roof penetrations: how many are acceptable (as drawn: four vents, three fan ducts, the dryer and the hood). Combining vents and routing fans to wall caps reduces it.
- Fire sprinklers: none shown. Say if the jurisdiction or the insurer wants them; the trunk route works for a multipurpose system.

Schedules

Fixture schedule

Tag	Fixture	Position	Drain	Trap	DFU	WSFU	Water	Stack
F1	Master WC toilet	4" from W23, 1'-7" from east ext. wall	3"	—	3	2.2	cold	S1
F2	Master vanity, left basin	5'-3" from W25, 9'-8" from north ext. wall	1-1/2"	1-1/4"	1	0.7	hot + cold	S1
F3	Master vanity, right basin	2'-3" from W25, 9'-8" from north ext. wall	1-1/2"	1-1/4"	1	0.7	hot + cold	S1
F4	Master tub, freestanding	1'-6" from north ext. wall, 6'-0" from east ext. wall	1-1/2"	1-1/2"	2	1.4	hot + cold	S1
F5	Roll-in shower, linear drain	3'-4" from W20, 4'-8" from north ext. wall	2"	2"	2	1.4	hot + cold	S1
F6	Powder toilet	6" from W20, 6'-2" from north ext. wall	3"	—	3	2.2	cold	S1
F7	Powder lav	11" from north ext. wall, 1'-3" from W20	1-1/2"	1-1/4"	1	0.7	hot + cold	S1
F8	Kitchen sink, island	9'-3" from W16, 12'-0" from west ext. wall	1-1/2"	1-1/2"	2	1.4	hot + cold	S2
F9	Dishwasher drawers	9'-3" from W16, 15'-6" from west ext. wall	to F8	—	—	1.4	—	S2
F10	Refrigerator water	10" from W16, 1'-4" from W15	—	—	—	0.5	cold	—
F11	Washer standpipe box	on W17, 2'-3" from W32	2"	2"	2	1.4	hot + cold	S3
F12	Laundry sink	1'-0" from W30, 1'-4" from W17	1-1/2"	1-1/2"	2	0.7	hot + cold	S3
F13	Suite tub / shower	1'-0" from W31, 2'-4" from W30	1-1/2"	1-1/2"	2	1.4	hot + cold	S3
F14	Suite toilet	4" from W7, 3'-9" from W31	3"	—	3	2.2	cold	S3
F15	Suite lav	10" from W30, 1'-6" from W9	1-1/2"	1-1/4"	1	0.7	hot + cold	S3
F16	Jack & Jill, left basin	3'-5" from W10, 6'-7" from south ext. wall	1-1/2"	1-1/4"	1	0.7	hot + cold	S4

Tag	Fixture	Position	Drain	Trap	DFU	WSFU	Water	Stack
F17	Jack & Jill, right basin	3'-5" from W11, 6'-7" from south ext. wall	1-1/2"	1-1/4"	1	0.7	hot + cold	S4
F18	Jack & Jill tub / shower	1'-5" from south ext. wall, 4'-8" from W10	1-1/2"	1-1/2"	2	1.4	hot + cold	S4
F19	Jack & Jill toilet	7" from south ext. wall, 1'-0" from W10	3"	—	3	2.2	cold	S4
F20	Frost-free sillcock, deck	on the west ext. wall, 1'-10" from W14	—	—	—	2.5	cold	—
F21	Floor drain, laundry	3'-5" from W31, 3'-5" from W30	2"	2"	2	0	—	S3
F22	Floor drain, master bedroom	6'-2" from W19, 9'-6" from east ext. wall	2"	2"	2	0	—	S1

Vent stacks and the drop

Stack	Position	Drain	Vent	In wall	DFU	Serves
S1	on W20, 9'-1" from north ext. wall	3"	3" VTR	W20 (2x6)	15	master WC, double vanity, tub, roll-in shower, powder toilet and lav, bedroom floor drain
S2	on W16, 3'-10" from W15	2"	2" VTR	W16	2	island sink and dishwasher (loop vent, P3112)
S3	on W30, 3'-11" from W17	3"	2" VTR	W30 (staggered)	12	washer, laundry sink and floor drain, suite tub, toilet and lav
S4	on W12, 9" from W10	3"	2" VTR	W12 (2x6)	7	Jack & Jill basins, tub and toilet
DR0P	4'-6" from west ext. wall, 6'-4" from W14	4"	—	floor	36	the single drop to the floor below, in the hallway by the deck door

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
PVC DWV 1-1/2" drain pipe	89 ft	+15%, fittings extra
PVC DWV 2" drain pipe	67 ft	+15%, fittings extra
PVC DWV 3" drain pipe	145 ft	+15%, fittings extra
PVC DWV 4" drain pipe	19 ft	+15%, fittings extra
PVC 1-1/2" vent pipe	43 ft	in-wall risers and ceiling-space runs
PVC 2" vent pipe	173 ft	in-wall risers and ceiling-space runs
PVC 3" vent pipe	16 ft	in-wall risers and ceiling-space runs
Roof vent flashings	4	one 3", three 2"; fewer if vents combine
3/4" PEX-A trunk, hot and cold	284 ft	plus 1/2" recirc return 142 ft
1/2" PEX-A branches, hot and cold	497 ft	

Item	Qty	Notes
1" PEX-A risers from the heater (R0)	40 ft	to confirm with the heater location
Stop valves, quarter-turn	33	one per supply
Hot-water recirculation demand pump	1	at the heater
Comfort-height elongated toilets, 1.28 gpf	4	
Lavatory basins (wall-hung vanities on S-1)	7	
Tubs with hand shower and pressure-balance valve	3	
Curbless shower: linear drain, thermostatic valve, slide-bar hand shower	1	
Kitchen sink, roll-under, with faucet	1	
Dishwasher drawer connection kit	1	
Refrigerator water box	1	
Washer outlet box with 2" standpipe	1	
Frost-free sillcock	1	
Floor drains 2" with trap primers	2	
Floor drop boot 4" through the floor, 4'-6" from west ext. wall, 6'-4" from W14	1	with cleanout above the floor
Cleanouts	6	base of each vent stack, the junction, and the drop
Rigid exhaust duct 4" / 6" with wall and roof caps	57 ft	8 runs (coordination)

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.